

ML Graded Quiz M3

Question 1

The choice of the splits in a decision tree is chosen such that _____.

- ☐ the Gini impurity is maximized and the overall information gain is minimized
- ☐ the MSE is maximized
- ☐ the Gini impurity is minimized and the overall information gain is maximized
- ☒ both Gini impurity and information gain are maximized

Question 2

Below, you'll see a table that shows a small sample of tweets that contain 3 categories of occurrence of a word against the sentiment of a certain topic. The phrase occurred either at a low, medium, or high frequency count, and the user's sentiment was confirmed as either being a positive sentiment (sentiment = 1) or negative sentiment (sentiment = 0).

phrase frequency (X)	Sentiment (Y)
High	0
High	0
Low	0
Med	0
Low	0
High	1
Low	1
Low	1
Low	1
Low	1
High	1
Med	1
Med	1
High	1

The prior probabilities for a negative and positive sentiment respectively are:

- ☐ $P(Y=0) = 0.733$ and $P(Y=1) = 0.844$
- ☐ $P(Y=0) = 0.643$ and $P(Y=1) = 0.357$
- ☐ $P(Y=0) = 0.733$ and $P(Y=1) = 0.844$
- ☒ $P(Y=0) = 0.357$ and $P(Y=1) = 0.643$

Question 3

Consider a sample of credit card clients where the default and non-defaulters are modeled against their income (X_2) and age (X_1) bracket.

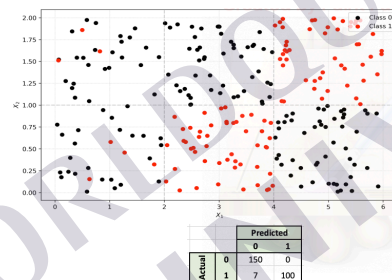
Based on this sample, what is the probability of default given a client earns \$4,000 and is 35 years old?

	X_1	X_1	X_2	X_2	Total
	< 30	30 & above	Income < \$2 500	Income >= \$2 500	Total
default ($Y = 1$)	175	100	120	155	275
non-default ($Y = 0$)	300	450	375	375	750
Total	475	550	495	530	1025

- ☐ 0.275
- ☐ 0.162
- ☐ 0.200
- ☒ 0.268

Question 4

Two classes of data are shown below in the feature space. Class 0 can be thought of as negatives with class 1 as positives. The confusion matrix for the classifier is shown below.



Which statement is **true** regarding the above information?

- ☐ The misclassifications are false negatives, as the tree most likely has a leaf node for class 0 such that $0 < X_1 < 2$.
- ☐ The misclassifications are false positives, as the tree most likely has a leaf node for class 0 such that $2 < X_1 < 3$.
- ☐ The misclassifications are false positives, as the tree most likely has a leaf node for class 0 such that $0 < X_1 < 2$.
- ☒ The misclassifications are false negatives, as the tree most likely has a leaf node for class 0 such that $2 < X_1 < 3$.

Question 5

Which adjustment of parameters can reduce over-fitting?

- ☐ Increasing maximum depth and increasing decreasing samples per leaf
- ☐ Decreasing the minimum weight fraction and minimum samples per leaf
- ☐ Increasing the minimum weight fraction and maximum depth

- ☒ Decreasing maximum depth and increasing minimum samples per leaf

Question 6

Below, you'll see a table that shows a sample of tweets that contain categories of occurrence of 2 words against the sentiment of a certain topic. The words occurred either at a low, medium, or high frequency count, and the user's sentiment was confirmed as either being a positive sentiment (sentiment = 1) or negative sentiment (sentiment = 0).

Word 1 count (X_1)	Word 2 count (X_2)	Sentiment (Y)
High	High	0
High	High	0
Low	High	0
Med	Low	0
Low	High	0
High	Low	1
Low	Low	1
Low	Low	1
Low	Low	1
Low	High	1
High	High	1
Med	Low	1
Med	Low	1
High	Low	1

What is the likelihood of $X_2 = \text{Low}$ given a positive sentiment?

- ☐ 0.8
- ☐ 0.125
- ☐ 0.20
- ☒ 0.778

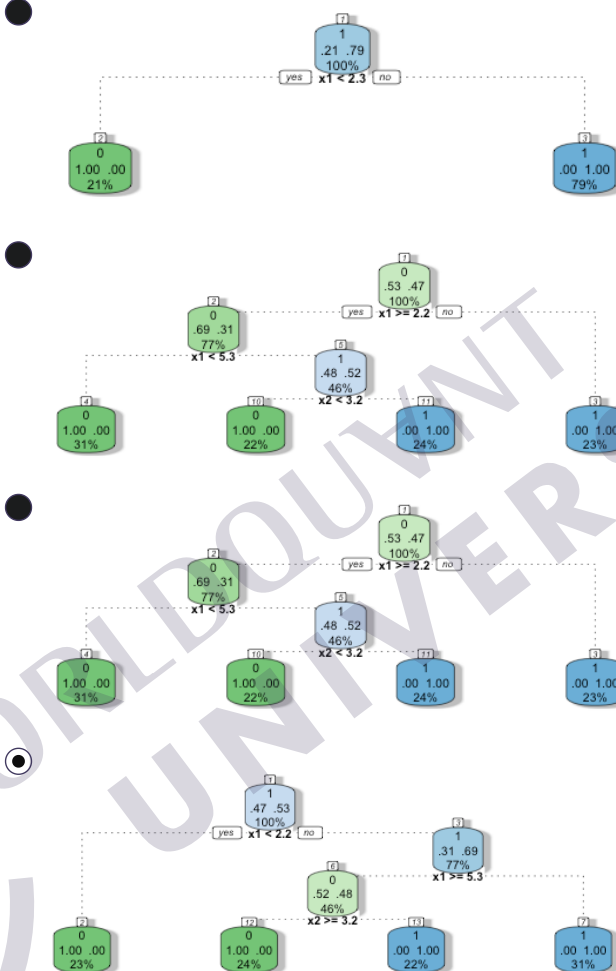
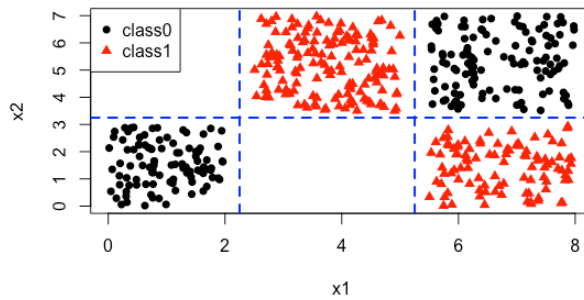
Question 7

Which of the below statements is **false**?

- ☐ The predicted target class requires the likelihood to be computed.
- ☐ The predicted target class does not require the Bayesian evidence to be computed.
- ☐ Prior probabilities for target classes are often taken to be the target class percentage.
- ☒ The predicted target class does not require the likelihood to be computed.

Question 8

Based on the data below, which tree best classifies the data?



Question 9

Which of the below is **true** about boosting models?

- ☐ It combines both decision trees and logistic regression.
- ☐ It is an algorithm built on trees in a parallel manner.
- ☐ It is an ensemble method built on weak decision trees.
- ☒ It is an extension of logistic regression.

Question 10

Below, you'll see a table that shows a sample of tweets that contain categories of occurrence of 2 words against the sentiment of a certain topic. The phrase occurred either at a low,

medium, or high frequency count, and the user's sentiment was confirmed as either being a positive sentiment (sentiment = 1) or negative sentiment (sentiment = 0).

Word 1 count (X_1)	Word 2 count (X_2)	Sentiment (Y)
High	High	0
High	High	0
Low	High	0
Med	Low	0
Low	High	0
High	Low	1
Low	Low	1
Low	Low	1
Low	Low	1
Low	High	1
High	High	1
Med	Low	1
Med	Low	1
High	Low	1

What is the likelihood of $X_1 = \text{Med}$ given a positive sentiment?

- ☐ 0.778
- ☐ 0.222
- ☐ 0.333
- ☒ 0.4

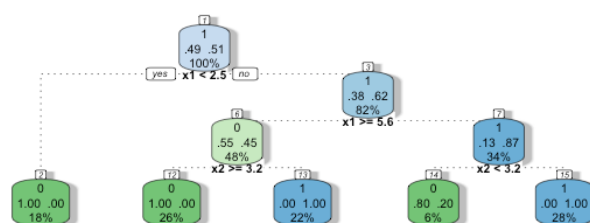
Question 11

Which of the following statements is **false**?

- ☐ Increasing the Bayesian evidence decreases the posterior probability of all target classes; however, this does not affect the Naive Bayes classification.
- ☐ Increasing the likelihood given target class k increases the posterior probability of that target class.
- ☐ Increasing the prior probability for target class k increases the posterior probability of that target class, hence increasing the chance of being classified as class k .
- ☒ Increasing the prior probability for target class k increases the posterior probability of that target class, hence decreasing the chance of being classified as class k .

Question 12

What is the highest Gini Impurity for the leaf nodes?



- ☐ 1
- ☐ 0.495
- ☐ 0.32
- ☒ 0.226

Question 13

Which of the following options is **true**?

- ☐ Logistic regression and decision trees cannot handle outliers and produce run-time errors when encountered.
- ☐ Logistic regression is affected by outliers whereas decision trees are not.
- ☐ Logistic regression and decision trees are not affected by outliers.
- ☒ Logistic regression and decision trees are affected by outliers.

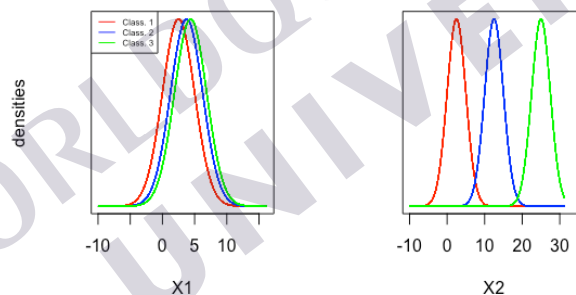
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QUESTIONS

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|----|----|----|----|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

Question 14

Likelihood densities of the two predictors with respect to the target classes are shown below.



- ☐ X1 is just as strong a predictor as X2.
- ☐ X1 is the weaker predictor of the target variable.
- ☐ X1 is a stronger predictor of the target variable.
- ☒ X1 is a strong predictor of Class 3.

Question 15

Returns that beat the market are classified as $Y=1$ with $Y=0$ otherwise. Suppose we have a Bayes table showing the likelihood of a predictor X below.

The predicted outcome of this is class 0, since the posterior is largest for $Y=0$.

Y	Prior	$P(X Y)$	Un_norm	Posterior	Predicted class
0	50%	0,33449	0,16724	0,696	0
1	50%	0,14643	0,07322	0,304	

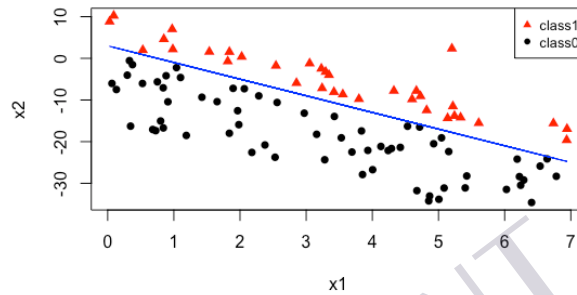
What is the posterior for $Y=1$?

- ☐ 1.000
- ☐ 0.2404
- ☐ 0.3045
- ☒ 0.4801

Question 16

The data below can be linearly separated by the blue line:

$$X_2 = 3 - 4X_1$$



Which would best classify class 0?

- ☐ $X_2 > 3 - 4X_1$
- ☐ $X_2 + 3 = -4X_1$
- ☐ $X_2 \leq 4X_1$
- ☒ $X_2 \leq 3 - 4X_1$

SUBMIT